

STATEMENT OF WORK

CREW SURVIVAL ENGINEERING (CSE) LABORATORY SERVICES

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1.0 GENERAL

1.1 Purpose

This statement of work (SOW) defines the tasks and deliverables associated with the Crew Survival Engineering (CSE) Laboratory services.

1.2 Background

The CSE Laboratory in Johnson Space Center's (JSC's) Crew and Thermal System Division (CTSD) is designing, developing, testing and integrating hardware as part of the Orion Crew Survival Systems (OCSS) project. The purpose of the OCSS project is to protect the crew in off nominal or contingency situations experienced during the mission that are life-threatening. All hardware in the CSE Laboratory is undergoing development per the OCSS project schedule for verification and validation testing, crew training, and Artemis missions. The flight hardware is estimated to be delivered to Orion for the Artemis missions in the December 2024 timeframe. Therefore, a significant amount of pre- and post-flight processing will be required in the CSE laboratory. The laboratory services shall support the following CTSD laboratory locations along with performing work in any contractor offsite facility:

- Building 7A/Rooms 336, 336A, 336B, 336AA, 336AB, and 336AC – Crew Survival Engineering (CSE) Laboratory[iesy](#)
- Building 7/Rooms 2020 and 2021 – Crew Survival Engineering (CSE) Laboratories

1.3 Scope of Effort

The Contractor shall provide the labor hours required to complete the effort defined in this SOW. This effort shall include the work described in Sections 2, 3, and 4 of this SOW.

2.0 REFERENCE AND APPLICABLE DOCUMENTS

The work performed shall conform to the requirements defined in this SOW and in the applicable documents listed below. Reference documents are listed below to assist in performing the scope.

2.1 Applicable Documents for CSE Laboratory

JSC 47745	Standard Operating Procedure for CELSE Laboratory
CTSD-SH-1289 (JSC 39185)	Integrated Hazard Analysis for the Crew Escape and Life Support Equipment Laboratory, Building 7A
CTSD-ADV-1001	Standard Operating Procedure for the Crew Escape Equipment Test Stand
CTSD-SH-1402 (JSC 47743)	Standard Operating Procedure for the CELSE Test Stand
CTSD-SH-2056	Hazard Analysis for the Life Support Equipment Test Stand Bldg 7A

CTSD-SH-2064	Modified Advanced Crew Escape Suit Testing Procedures
CTSD-SH-2067	Hazard Analysis for Advanced Crew Escape Suit (ACES) & Modified ACES Testing/Evaluations
ES4-10-042A	Operation and Configuration Control Plan (OCCP) for Safe Ground Pressurization of Advanced Crew Escape Suit
JSC-47101	OCSS Project Technical Requirements Specification

2.2 Reference Documents for CSE Laboratory

N/A

3.0 CSE TASK DESCRIPTIONS

The following section provides a description of the scope of work for CSE Laboratory services. This scope includes the services of ~~1000-1050~~ **hours** of a CSE-qualified Laboratory technician (**1000 hours**) as well as project management oversight (50 hours). The Contracting Officer's Representative (COR) shall communicate detailed tasks within the description of this SOW.

3.1 CSE Laboratory Technician Services

The contractor shall provide **1000 hours** of a CSE-qualified laboratory technician to service the CSE Laboratory daily operations. The work environment shall encompass any of the building 7 and building 7A laboratories or at the contractor's offsite facilities to support the project. The specific laboratory services shall be performed in the CSE Laboratory (i. e. Building 7A/Rooms 336, 336A, 336B, 336AA, 336AB, and 336AC and Building 7/Rooms 2020 and 2021).

The contractor shall provide laboratory services for tasks such as follows:

- a. CSE Laboratory testing with hardware such as the MACES/ACES/OCSS suits, Orion crew survival hardware, or additional OCSS project systems;
- b. Cable fabrication;
- c. Calibration;
- d. General maintenance;
- e. Laboratory readiness;
- f. Engineering evaluation tests;
- g. Test operation including start up and shut down of tests;
- h. Test rig buildup and take down;
- i. Writing of Test Preparation Sheets (TPS's);
- j. Buildup of mockups and ground test equipment; and
- k. Specialized space suit mechanical and electrical assembly testing.

3.2 CSE Laboratory Project Management

The contractor shall provide 50 hours of project management oversight for the laboratory services provided.

4.0 DELIVERABLES

The contractor shall deliver data and hardware as described below:

No.	Title	Comments	Quantity	Due
1	Monthly Status	A summary report shall be prepared by the contractor in Word or PowerPoint equivalent. The report shall itemize monthly activities with key information such as metrology and calibration status along with hours used and cost expended.	1 per month	2 nd Monday each Month after authority to proceed (ATP)
2	Lessons Learned	Presentation of lessons learned	A/R	A/R